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Identifying critical transmission lines in cascading outages using the natural connectivity index

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Abstract
This paper presents a novel method for identifying critical transmission lines (CTLs) in power systems subjected to cascading overload outages, leveraging the Natural Connectivity (NC) index. Unlike previous studies that employed suboptimal weights for the edges, this work introduces a new weighting strategy that enhances the effectiveness of NC in CTLs identification. The proposed Weighted Natural Connectivity-based (WNC-based) method is validated through comparisons with established methods, including the catastrophic expectation index, and the cascading failure simulation index, using IEEE test systems of 30, 39, 57, and 118 buses and the 400 kV Mexican power system. The experimental results demonstrate that the WNC-based method accurately identifies critical lines, providing a practical balance between simplicity and precision. This study challenges prior conclusions that natural connectivity is unsuitable for CTLs identification, showcasing its potential in power system applications. Furthermore, the method's applicability extends to interconnected systems, such as the integration of power transmission, gas distribution, and telecommunication networks, paving the way for future research. The WNC-based method offers a practical tool for preventing blackouts and enhancing power grid resilience, contributing significantly to the field of power system reliability and stability.

Keywords Cascading · Complex · Identification · Lines · Vulnerable

1 Introduction

Modern power systems are becoming increasingly complex, which places significant demands on their safety and reliability. One of the most pressing issues in recent years has been cascading outages, which lead to extensive load shedding and significant social impacts [1]. In these outages, critical transmission lines (CTLs) often act as triggers for the spread of failures. For example, in July 2017, a downed transmission line in Panama caused power disruptions across Central America, affecting millions [2]. Similarly, in March 2018, a major outage in Brazil was triggered by the failure of a transmission line near the Belo Monte hydroelectric station, impacting tens of millions [3]. In August 2019, a blackout in Indonesia affected 21.9 million customers after an issue in a 500 kV transmission line [4]. More recently, in December 2020, an outage in Mexico affected 10.3 million consumers, caused by the shutdown of two 400 kV transmission lines [5].

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These incidents demonstrate the critical role of certain transmission lines in the structure of power grids. When these key lines fail, the robustness of power grids against cascading outages is greatly compromised. Therefore, identifying CTLs has become a focus in research to prevent cascading failures and improve grid reliability.

Methods to identify CTLs generally fall into three categories: simulation-based approaches, topological methods, and complex network theory incorporating electrical characteristics. In simulation-based methods, several approaches have been proposed. For instance, an incremental power flow entropy model was developed that considers branch power flow transfer and distribution entropy [6]. A line risk index was proposed based on line overload probability and power flow fluctuations [7]. Other methods include finding power components expected to cause large-scale cascading failures [8] and identifying sensitive lines based on system cascading collapses [9]. Cascading failures have also been simulated using fault chain theory, where the number of disconnected lines is used to determine a vulnerability index [10].

In topological structure methods, approaches such as using topological potential have been developed to identify vulnerable lines, including consideration of wind power uncertainty [11]. A modified PageRank method was proposed that includes power transfer and security margin after an N-1 contingency check [12]. Another method based on improved structural hole theory, considering changes in topology after N-1 checks, was proposed to identify vulnerable lines [13, 14].

Complex network theory, combined with electrical characteristics, is another approach for identifying CTLs. One method proposes a novel betweenness index that uses transmission line reactance as edge weights in the network [15]. Other works use integrated "betweenness" indices [16] or electrical betweenness calculated by injecting unit current between source and sink [17]. Network centrality, based on topological and system response features, has also been introduced [18], along with K-shell decomposition [19].

Recently, additional methods that do not fit into these categories have been proposed. For example, a method based on a weighted H-index was developed to identify vulnerable lines by reflecting the correlation between transmission branches in active power transmission and constructing a correlation network [20]. Another method was introduced to identify weak lines in power systems with intermittent energy resources and DC line integration [21]. Additionally, a method based on game theory, which incorporates network topology and operational states, was proposed for identifying vulnerable lines [22].

However, despite the contributions of these methods, certain challenges remain. Some approaches require detailed system data that may not be available due to security concerns. Others are too complex for practical use in real-world evaluations or involve computationally intensive simulations, particularly for large-scale power grids.

To address these challenges, this paper proposes a novel method for identifying CTLs using the Natural Connectivity Index (NCI). Power grids typically have multiple pathways between nodes, allowing continued operation even if one path fails. The more paths a grid has, the more robust it is. NCI, introduced in ref. [23], is a structural robustness index based on the Estrada index [24] that quantifies the number of closed walks of all lengths in the network. Derived from the graph's eigenvalues, NCI provides valuable insights into the topology and dynamics of complex networks [25–28].

Although NCI has been used before to identify critical lines [20, 34], previous studies found it ineffective due to the weight assignment strategies employed. The core of the method proposed here is a new weight assignment strategy that makes NCI effective for identifying critical transmission lines.

The key contributions of this work are as follows: (a) A novel method, referred to as the WNC-based method, which utilizes the NCI specifically adapted for weighted graphs to identify CTLs, demonstrating improved results compared to previous findings in the literature. (b) The WNC-based method does not require sensitive system data, making it more applicable in situations where such data is unavailable. (c) The method is simple, relying on the adjacency matrix and eigenvalues of the graph, and can be practically applied to various power systems.

The remainder of this paper is organised as follows: Sect. 2 provides an overview of the Natural Connectivity index. Section 3 describes the cascading failure simulator used in this study. Section 4 introduces the proposed WNC-based method. Section 5 presents case studies applying the WNC-based method to various IEEE test systems and a real-life power grid, validating its effectiveness through comparisons with other established methods, such as the Catastrophic Expectation Index (CEI), the Cascading Failure Simulation (CFS) index, and an AC cascading outage simulator. Finally, Section 6 concludes with a summary of the key results and suggestions for future research directions.

2 Natural connectivity

2.1 Natural connectivity for simple unweighted graphs

A power grid can be represented by an undirected graph $\boldsymbol{G} = (\boldsymbol{V}, \boldsymbol{E})$ where V is the set of vertices (or nodes) and E is the set of edges. The number of nodes in G is denoted by N. Let $\lambda_1 \geq \lambda_2 \geq ... \geq \lambda_{\boldsymbol{N}}$ represent the eigenvalues of the adjacency matrix A of G, commonly referred to as the eigenvalues of graph G. The spectrum of G is composed of $\{\lambda_1, \lambda_2, ..., \lambda_{\boldsymbol{N}}\}$.

A walk of length k in a graph G is an alternating sequence of nodes and edges $v_0e_1v_1e_2...v_ke_k$, where $v_i \in V$ and $e_i \in E$. A walk is closed if $v_0 = v_k$. The number of closed walks is a key index for complex networks. The number of closed walks of all lengths quantifies the redundancy of alternative paths in the graph and can be used to assess the graph's robustness. Since shorter closed walks have a greater impact on redundancy than longer ones, their contribution is weighted by dividing them by the factorial of their length k. That is, $S = \sum_{k=0}^{\infty} \frac{n_k}{k!}$, where n_k is the number of closed walks of length k. This ensures that the weighted sum does not diverge and allows S to be calculated using the powers of the adjacency matrix:

$$n_k = \sum_{i=1}^{N} \lambda_i^k = Trace\left(A^k\right) \tag{1}$$

Therefore,

$$S = \sum_{k=0}^{\infty} \frac{n_k}{k!} = \sum_{k=0}^{\infty} \sum_{i=1}^{N} \frac{\lambda_i^k}{k!} = \sum_{i=1}^{N} \sum_{k=0}^{\infty} \frac{\lambda_i^k}{k!} = \sum_{i=1}^{N} e^{\lambda_i} \tag{2}$$

The larger the value of S, the greater the number of redundant alternative paths in the graph, resulting in better robustness. It is important to note that (2) corresponds to the Estrada index of the graph [24], which has been applied in various contexts in graph theory.

The NCI of G is defined as [23]:

$$\overline{\lambda}(G) = ln\left(\frac{S}{N}\right) = ln\left(\frac{1}{N}\sum_{i=1}^{N} e^{\lambda_i}\right) \tag{3}$$

which is equivalent to an "average" eigenvalue of G since $\lambda_N \leq \overline{\lambda} \leq \lambda_1$. $\overline{\lambda}$ represents the natural logarithm of the density average of the graph's characteristic spectrum.

The maximum NCI for a graph with N nodes is obtained in a complete graph, which is $N - lnN$ as $N \to \infty$. To compare the robustness of graphs of different sizes, NCI is divided by the maximum natural connectivity $N - lnN$.

2.2 Natural connectivity for weighted graphs

For a graph G with edge weights $w_{ij} \in \mathbb{R}^+$, $W = \left(w_{ij}\right)_{N\times N}$ is the weighted adjacency matrix of G, and the number of closed walks of length k is defined as:

$$n_k = \sum_{i=1}^{N} \lambda_i^k = Trace\left(W^k\right) \tag{4}$$

The NCI for weighted graphs $\overline{\lambda}(W)$ is then obtained from the corresponding eigenvalues of the matrix W [29]. In cases where some edges have excessive influence due to higher weights, Crofts and Higham [30] proposed normalising the weighted adjacency matrix as follows: $\widetilde{W} = K^{-1/2}WK^{-1/2}$, where K is a diagonal matrix of weighted degrees.

3 Cascading outage simulator

This section describes the cascading outage simulator presented in ref. [31], which is used to validate the results of the proposed method. The simulator is implemented through the following steps:

1. Select and disconnect overloaded components after a contingency occurs.

2. Update the system (generators, lines, and loads).
3. If the system becomes divided into islands, balance the generation and load in each island by shedding the load or adjusting the generator outputs.
4. Compute AC power flow in each island.
5. If there are components violating their capacity limits, return to Step 1; otherwise, proceed to Step 6.
6. Calculate load losses or other cascade size metrics and stop the simulation.

In Step 3, if the total supply of active power within an island surpasses the total demand, the active power outputs of the generators are curtailed. Conversely, if the total supply of active power falls short of satisfying the total demand, demand reduction is employed to balance supply and demand within the island. This involves scaling down the energy supply or demand across all nodes by a common factor. In the event that an island lacks both supply and demand, all supply and demand nodes within the island are deactivated. For each island containing both supply and demand nodes, the cascading outage simulator designates the slack node as the controlled voltage node with the maximum power output in that component.

Additionally, a deterministic rule is used to determine transmission line outages. According to this rule, a line fails when the power flow magnitude on that line, denoted by $|f_l|$, exceeds its capacity c_l. The magnitude $|f_l|$ of flow on a line l_{ij} is defined as.

$$|f_l| = \frac{|S_{ij}| + |S_{ji}|}{2}, \tag{5}$$

where S_{ij} and S_{ji} are the sent and received apparent power flows on the line l_{ij}.

In some test networks, the maximum line flow capacities are not provided. In such cases, the maximum power flow that a transmission line can tolerate is defined as its maximum capacity. Additionally, thermal, stability, and voltage drop constraints impose flow limits on transmission lines. In real-world infrastructures, the maximum capacity of a transmission line is constrained by cost [32]. Following the methodology in ref. [31], line flow capacities are estimated as $c_l = (1+\alpha) max\left\{|f_l|, \bar{f}\right\}$, where α is the line tolerance and $\bar{f}$ is the mean of the initial magnitudes of line flows.

The metric used to measure the severity of a cascade, which is employed to identify CTLs, is the ratio of the demand supplied at the end of the cascade to the initial demand. This ratio is denoted by Y_g.

4 The proposed method

The core of the proposed WNC-based method involves calculating the weights w_{ij} for each edge, representing each transmission line l_{ij} of a power grid as follows:

$$w_{ij} = \frac{|f_l|}{|f_{max}|}, \tag{6}$$

where $|f_l|$ is the power flow magnitude of line l_{ij} and $|f_{max}|$ is the maximum power flow transmitted in any transmission line of the power grid.

The theoretical basis for this approach is rooted in the observed global correlation effects in power systems. Cascades of local failures in power grids have often resulted in mass blackouts, implying that the failure of any non-radial line is linked to nearly every other line through the network as a whole, rather than through individual lines. Furthermore, in power flow analysis, the slack bus serves as the reference bus with an angle typically set to zero degrees and a voltage magnitude near 1.0. The voltage magnitude and phase angle at each bus are computed relative to the slack bus. Based on this, the hypothesis of this work is that there is a correlation between the weights of all edges, and one way to establish this correlation is by using the ratio of the power flow in each transmission line to the maximum power flow in the grid.

The proposed WNC-based method for identifying CTLs in power grids consists of the following eight steps:

Step 1: Select the power grid to be analysed.

Step 2: Solve the pre-contingency power flow to obtain the line flows $|f_l|$ and maximum line flow capacities c_l for the undamaged power grid.

Step 3: Calculate the edge weights w_{ij} and construct the undamaged weighted adjacency matrix W.

Step 4: Compute the eigenvalues of W and use them to calculate the natural connectivity $\overline{\lambda}_{initial}$ for the undamaged power grid.

Step 5: Remove line l from W and compute the natural connectivity $\overline{\lambda}_{E-1}$ for the damaged power grid.

Step 6: Evaluate the robustness to the analysed contingency as follows:

$$\overline{\lambda}_r = \frac{\overline{\lambda}_{initial} - \overline{\lambda}_{E-1}}{\overline{\lambda}_{initial}}. \tag{7}$$

Step 7: Repeat steps 5 and 6 until all the single-line failures are analysed.

Step 8: Identify the critical transmission lines with the highest $\overline{\lambda}_r$.

5 Case studies

In this section, the CTLs of five synthetic realistic power systems are identified using the proposed WNC-based method. The systems analysed include the IEEE 24-bus, IEEE 30-bus, IEEE 39-bus, IEEE 57-bus, and IEEE 118-bus test systems [33]. To validate the effectiveness of the proposed method, the results obtained are compared with those reported in refs. [20, 31, 34], which use three cascading indices to identify CTLs: the Y_g index, the catastrophic expectation index (CEI) [35], and the cascading failure simulation (CFS) index [12].

To identify the CTLs using the Y_g index, the cascading outage model presented in ref. [31] and described in Sect. 3 was employed. This model, implemented in MATPOWER [36], is publicly available in ref. [37]. For each cascading outage simulation initiated by a single-line failure, the decline in the PG's performance is measured by the Y_g index. Similarly, for each single-line failure, the relative decrease in the natural connectivity index $\overline{\lambda}_r$ is calculated using (7). A smaller value of Y_g indicates a more critical line, while a larger value of $\overline{\lambda}_r$ suggests a more critical line.

5.1 Validating the WNC-based method with the CEI and CFS indices

In this subsection, the IEEE 24-bus reliability test system and the IEEE 39-bus system are analysed. The CTLs identified by the proposed WNC-based method are compared with those identified by the CEI and CFS indices, as reported in refs. [20, 34], respectively. The CEI assesses the cascading overload risk by estimating the probability of a cascading trip on line j following the outage of line i. The CFS index considers the incorrect tripping probability of each line after a transmission line is randomly tripped. After T time simulations, the average load loss following each line disconnection is calculated. If the WNC-based method identifies cln lines that match the lines identified by the CEI and CFS (denoted as M), the identification accuracy is defined as cln/M.

Table 1 shows the critical lines of the IEEE 24-bus reliability test system identified by the WNC-based method ($\overline{\lambda}_r$) and the CEI index reported in ref. [34]. Both methods identified line 25 as the most critical. Additionally, the WNC-based method identified three more critical lines (27, 7, and 36) that were also identified by the CEI index among the five most

Table 1 Most CTLs of the IEEE 24-bus system

Rank	CEI index [34]	WNC-based method
1	25	25
2	38	23
3	27	28
4	7	21
5	36	22
6	–	27
7	–	36
8	–	7
cln/M		4/5

Table 2 Most CTLs of the IEEE 39-bus system

Rank	CFS index [20]	WNC-based method
1	46	46
2	38	14
3	35	37
4	14	20
5	28	33
6	20	35
7	37	10
8	18	39
cln/M		5/8

Table 3 Most CTLs of the IEEE 30-bus system

Rank	Cascading outage simulator Y_g	WNC-based method		
		$w_{ij} = 1$	$w_{ij} = \lvert f_l \rvert / \lvert f_{max} \rvert$	$w_{ij} = PR_l$
1	12–13	6–10	12–13	2–6
2	21–22	2–6	6–8	2–4
3	6–8	4–6	4–6	4–6
4	1–2	6–9	21–22	10–21
5	4–6	2–4	2–6	21–22
cln/M		1/5	4/5	2/5

critical lines, resulting in an identification accuracy of 80%. In ref. [34], natural connectivity for weighted graphs was used to validate the CEI index. The authors concluded that natural connectivity could not identify critical lines for the IEEE 24-bus system. However, using the proposed weight assignment in this work, natural connectivity successfully identified critical lines for this system, contradicting the conclusions in ref. [34].

Table 2 lists the critical lines of the IEEE 39-bus system identified by the proposed WNC-based method and the CFS index reported in ref. [20]. Both methods identified line 46 as the most critical. Additionally, the WNC-based method identified six more critical lines that matched some of the critical lines identified by the CFS index among the twelve most critical lines, resulting in an identification accuracy of 62.5%. In ref. [20], natural connectivity for weighted graphs was also used to validate an approach based on the weighted H-index. The results in ref. [20] indicated that neither natural connectivity nor the weighted H-index method could identify line 46, which the CFS index had identified as the most critical. However, the proposed WNC-based method successfully identified this line, outperforming the weighted H-index method proposed in ref. [20].

In both cases, the results obtained by the WNC-based method using the proposed weights contradict the conclusions in refs. [20, 34], which claimed that natural connectivity is unsuitable for identifying CTLs in power systems.

5.2 Validating the WNC-based method with an AC cascading outage simulator

The previous indices, CEI and CFS, utilise the DC power flow equations, which are based on a linearisation of the non-linear AC power flow dynamics. The induced linearisation error may be minimal in large transmission grids but can be significant for certain PGs [38]. Motivated by these observations, the proposed WNC-based method was validated using the AC cascading outage simulator presented in ref. [31] and available in ref. [37]. The IEEE 30-bus and IEEE 118-bus test systems were employed for this validation.

Tables 3 and 4 show the CTLs identified by the Y_g index of the AC cascading outage simulator and by the WNC-based method with three different weight assignments for w_{ij}: a constant value of 1.0, the proposed weight from (6), and the percentage of rating (PR_l). The second column in each table presents the results sorted by the Y_g index. To evaluate the performance of the proposed method more rigorously, only the first five most critical lines identified by the two methods

Table 4 Most CTLs of the IEEE 118-bus system

Rank	Cascading outage simulator Y_g	WNC-based method		
		$w_{ij} = 1$	$w_{ij} = \|f_l\|/\|f_{max}\|$	$w_{ij} = PR_l$
1	30–17	54–59	8–9	92–94
2	8–9	56–59	9–10	5–11
3	9–10	69–75	8–5	69–75
4	8–5	54–56	49–66	100–103
5	89–90	92–100	37–38	60–61
cln/M		0/5	3/5	0/5

were considered. The proposed WNC-based method, using the weight assignment strategy from (6) (fourth column), was able to identify four of the five most critical lines identified by the Y_g index in the IEEE 30-bus, and three of the five most critical lines in the IEEE 118-bus system. These results demonstrate that the critical lines in cascading outages are influenced not only by the power flow but also by the topology of the power grids. Specifically, they show that the number of parallel paths between pairs of nodes plays a key role in determining the robustness of the grid.

5.3 Validating the WNC-based method with a real-life power grid

To further validate the effectiveness of the WNC-based method on real-life power systems, an analysis of the 400 kV power grid of Mexico, consisting of 169 transmission lines and 127 nodes (see Fig. 1), was performed. Table 5 presents

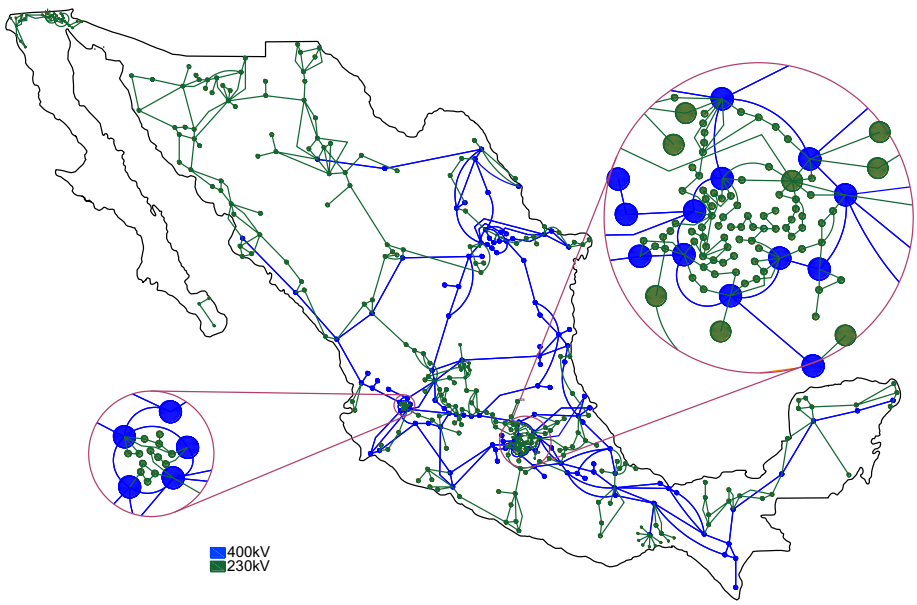


Fig. 1 Power grid of Mexico at 230 kV and 400 kV

Table 5 Most CTLs of the Mexican 400 kV power grid

Rank	Cascading outage simulator Y_g	WNC-based method
1	EPS-400, ANP-400	PAE-400, CPY-400
2	TUL-400, VIC-400	TEX-400, TUV-400
3	TEX-400, TUV-400	QPM-400, MES-400
4	LAP-400, TEX-400	LAP-400, TEX-400
5	PAE-400, CPY-400	ALT-400, CPY-400
cln/M		3/5

a comparison of the identification results of the proposed WNC-based method with the Y_g index of the AC cascading outage simulator.

As shown in Table 5, the WNC-based method aligns with the Y_g index in identifying three out of the five most CTLs. These three lines are crucial in this power grid because they are located in two regions with high electricity consumption: the central and northeast regions. Additionally, power flow congestion has been reported in the neighboring lines of two of these three critical lines [39].

6 Conclusions

This paper proposed a novel WNC-based method for identifying critical transmission lines (CTLs) in power grids, using the natural connectivity index (NCI) with a new weight assignment strategy. The motivation for this research was rooted in the limitations of previous methods, such as the need for extensive operational data and computational complexity. The proposed method addressed these challenges by offering a more efficient, robust, and practical approach that correlates power flow in transmission lines with the overall robustness of the power grid structure.

The method was applied to both synthetic and real-world power grids. In the case studies involving the IEEE 24-bus, 30-bus, 39-bus, 57-bus, and 118-bus systems, the results of the proposed WNC-based method were compared with those obtained from well-established indices, including the catastrophic expectation index (CEI), the cascading failure simulation (CFS) index, and the Y_g index from an AC cascading outage simulator. The comparison revealed that the WNC-based method identified critical transmission lines with a high degree of accuracy, closely matching the results of these established indices. In some instances, such as the IEEE 24-bus and IEEE 39-bus systems, the WNC-based method demonstrated superior accuracy by successfully identifying critical lines that were missed by other methods.

However, there are some limitations to the proposed method. One notable limitation is that it relies on the accuracy of the initial power flow data, which may not always be readily available in certain power grids. Additionally, while the method is computationally efficient for medium-sized grids, further improvements are necessary to ensure scalability for extremely large and interconnected systems. Future research could focus on refining the weight assignment strategy to account for dynamic factors, such as fluctuating loads and renewable energy integration, which may affect grid robustness in real time.

Furthermore, the hardware implementation of the proposed scheme is both possible and significant. The simplicity of the method's reliance on matrix calculations and the use of eigenvalues makes it suitable for real-time applications in control systems and hardware, especially for monitoring and mitigating cascading failures in power systems. By integrating this method into hardware solutions such as supervisory control and data acquisition (SCADA) systems or real-time digital simulators (RTDS), grid operators could gain the ability to identify and respond to critical line failures more rapidly, thus improving grid resilience. The ease of hardware implementation also enhances the practicality of the proposed method for use in modern smart grids, where real-time monitoring and fast decision-making are essential.

From a societal perspective, the method presented in this paper has significant implications for enhancing the reliability and resilience of power grids, which are essential infrastructure systems. By effectively identifying critical transmission lines, the proposed method can help prevent large-scale blackouts, which have severe social and economic consequences. Reliable power supply is vital for the functioning of modern societies, and disruptions can lead to critical failures in healthcare, transportation, communication, and other key services. The ability to identify and mitigate cascading failures in power grids contributes directly to the stability of these services, improving the overall quality of life and promoting social and economic development.

Moreover, the effectiveness of the WNC-based method was further demonstrated through validation with the AC cascading outage simulator on the IEEE 30-bus and IEEE 118-bus systems, where the method successfully identified four of the five most critical lines in the former and three of the five most critical lines in the latter. Additionally, the analysis of Mexico's 400 kV power grid highlighted the method's applicability to real-life power systems, particularly in regions with high electricity consumption. The consistency of the results with previously reported power flow congestion issues further reinforces the method's practical relevance for both synthetic and real-world applications.

The key contributions of this research are:

- The development of a novel weight assignment strategy that enhances the applicability of the natural connectivity index for identifying CTLs.

- The validation of the WNC-based method across a range of synthetic and real-world power systems, demonstrating its robustness and practicality.
- The proposed method closely matches or exceeds the performance of established cascading failure indices, while offering a simpler and more computationally efficient approach.
- The potential for hardware implementation, enabling real-time monitoring and enhanced decision-making capabilities for grid operators.
- The societal relevance of the proposed method, which can contribute to the prevention of large-scale blackouts and enhance the reliability of power grids, benefiting modern societies by ensuring continuous and reliable access to electricity.

In conclusion, the WNC-based method provides an effective and reliable tool for identifying critical transmission lines in power systems. Its ability to overcome the limitations of previous methods and its validation through both synthetic and real-life case studies confirm its potential for real-world applications in enhancing the robustness and reliability of power grids. Future research may focus on extending the method to interconnected systems, improving scalability, and further exploring its hardware integration to mitigate cascading failures in increasingly complex power grids.

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Data availability The datasets generated during and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Competing interests The authors declare no competing interests.

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References

1. Guo H, Zheng C, Iu HHC, Fernando T. A critical review of cascading failure analysis and modeling of power system. Renew Sustain Energy Rev. 2017;80:9–22. https://doi.org/10.1016/j.rser.2017.05.206.
2. AFP. Central America hit by massive power outages. 2017. https://sg.news.yahoo.com/massive-power-outage-hits-central-america-235450509.html/. Accessed 4 Jan 2021.
3. U.S. News Online. Tens of millions in northern Brazil Hit by massive power outage. 2018. https://www.usnews.com/news/world/articles/2018-03-21/tens-of-millions-in-northern-brazil-hit-by-massive-power-outage/. Accessed 4 Jan 2021.
4. The Japan Times Online. Major blackout darkens Jakarta, neighbouring provinces on Java. 2019. https://www.japantimes.co.jp/news/2019/08/04/asia-pacific/major-blackout-darkens-jakarta-neighboring-provinces-java/. Accessed 4 Jan 2021.
5. The Yucatan Times. Millions without light after massive blackout in Mexico. 2020. https://www.theyucatantimes.com/2020/12/millions-without-light-after-massive-blackout-in-mexico/. Accessed 4 Jan 2021.
6. Fang R, Shang R, Wang Y, Guo X. Identification of vulnerable lines in power grids with wind power integration based on a weighted entropy analysis method. Int J Hydrogen Energy. 2017;42(31):20269–76. https://doi.org/10.1016/j.ijhydene.2017.06.039.
7. Li X, Zhang X, Wu L, Lu P, Zhang S. Transmission line overload risk assessment for power systems with wind and load-power generation correlation. IEEE Trans Smart Grid. 2015;6(3):1233–42. https://doi.org/10.1109/TSG.2014.2387281.
8. Eppstein MJ, Hines PD. A "random chemistry" algorithm for identifying collections of multiple contingencies that initiate cascading failure. IEEE Trans Power Syst. 2012;27(3):1698–705. https://doi.org/10.1109/TPWRS.2012.2183624.

9. Salim NA, Othman MM, Musirin I, Serwan MS, Busan S. Risk assessment of dynamic system cascading collapse for determining the sensitive transmission lines and severity of total loading conditions. Reliab Eng Syst Saf. 2017;157:113–28. https://doi.org/10.1016/j.ress.2016.08.002.
10. Wang A, Luo Y, Tu G, Liu P. Vulnerability assessment scheme for power system transmission networks based on the fault chain theory. IEEE Trans Power Syst. 2010;26(1):442–50. https://doi.org/10.1109/TPWRS.2010.2052291.
11. Zhang L, Zhu B, Wang Y. Identification of vulnerable lines in power grids with wind power integration based on topological potential. Electr Power Syst Res. 2024;234:110593. https://doi.org/10.1016/j.epsr.2024.110593.
12. Ma Z, Shen C, Liu F, Mei S. Fast screening of vulnerable transmission lines in power grids: a pagerank-based approach. IEEE Trans Smart Grid. 2017;10(2):1982–91. https://doi.org/10.1109/TSG.2017.2785267.
13. Wenli F, Xuemin Z, Shengwei M, Shaowei H, Wei W, Lijie D. Vulnerable transmission line identification using ISH theory in power grids. IET Gener Transm Distrib. 2018;12(4):1014–20. https://doi.org/10.1049/iet-gtd.2017.0571.
14. Hu P, Fan WL. Mitigation strategy against cascading failures considering vulnerable transmission line in power grid. Phys A. 2020;540:123230. https://doi.org/10.1016/j.physa.2019.123230.
15. Dwivedi A, Yu X, Sokolowski P. Identifying vulnerable lines in a power network using complex network theory. In: 2009 IEEE International Symposium on Industrial Electronics. IEEE; 2009. pp. 18–23. https://doi.org/10.1109/ISIE.2009.5214082.
16. Wang Z, He J, Nechifor A, Zhang D, Crossley P. Identification of critical transmission lines in complex power networks. Energies. 2017;10(9):1294. https://doi.org/10.3390/en10091294.
17. Xu L, Wang XL, Wang XF. Electric betweenness and its application in vulnerable line identification in power system. Proc CSEE. 2010;30(1):33–9.
18. Song H, Dosano RD, Lee B. Power grid node and line delta centrality measures for selection of critical lines in terms of blackouts with cascading failures. Int J Innov Comput Inf Control. 2011;7(3):1321–30.
19. Fan WL, Zhang XM, Mei SW, Huang SW. Vulnerable transmission line identification considering depth of K-shell decomposition in complex grids. IET Gener Transm Distrib. 2018;12(5):1137–44. https://doi.org/10.1049/iet-gtd.2017.0767.
20. Zhang Q, Fan W, Qiu Z, Liu Z, Zhang J. A new identification approach of power system vulnerable lines based on weighed H-index. IEEE Access. 2019;7:121421–31. https://doi.org/10.1109/ACCESS.2019.2937903.
21. Gao Q, Wang Y, Cheng X, Yu J, Chen X, Jing T. Identification of vulnerable lines in smart grid systems based on affinity propagation clustering. IEEE Internet Things J. 2019;6(3):5163–71. https://doi.org/10.1109/JIOT.2019.2897434.
22. Zhu D, Wang R, Duan J, Cheng W. Comprehensive weight method based on game theory for identify critical transmission lines in power system. Int J Electr Power Energy Syst. 2021;124:106362. https://doi.org/10.1016/j.ijepes.2020.106362.
23. Jun WU, Barahona M, Yue-Jin T, Hong-Zhong D. Natural connectivity of complex networks. Chin Phys Lett. 2010;27(7): 078902. https://doi.org/10.1088/0256-307X/27/7/078902.
24. De La Peña JA, Gutman I, Rada J. Estimating the Estrada index. Linear Algebra Appl. 2007;427(1):70–6. https://doi.org/10.1016/j.laa.2007.06.020.
25. Peng GS, Wu J. Optimal network topology for structural robustness based on natural connectivity. Phys A. 2016;443:212–20. https://doi.org/10.1016/j.physa.2015.09.023.
26. Shen A, Guo J, Wang Z. Research on methods for improving robustness of cascading failures of interdependent networks. Wirel Pers Commun. 2017;95:2111–26. https://doi.org/10.1007/s11277-017-4041-6.
27. Pizzuti C, Socievole A, Van Mieghem P. Comparative network robustness evaluation of link attacks. In: Complex networks and their applications VIII: Volume 1 Proceedings of the Eighth International Conference on Complex Networks and Their Applications COMPLEX NETWORKS. 2019. Springer International Publishing; 2020. pp. 735–46. https://doi.org/10.1007/978-3-030-36687-2_61.
28. Chan H, Akoglu L. Optimizing network robustness by edge rewiring: a general framework. Data Min Knowl Discov. 2016;30(5):1395–425. https://doi.org/10.1007/s10618-015-0447-5.
29. Xiao-Ke Z, Jun W, Yue-Jin T, Hong-Zhong D, Yong L. Structural robustness of weighted complex networks based on natural connectivity. Chin Phys Lett. 2013;30(10):108901. https://doi.org/10.1088/0256-307X/30/10/108901.
30. Crofts JJ, Higham DJ. A weighted communicability measure applied to complex brain networks. J R Soc Interface. 2009;6(33):411–4. https://doi.org/10.1098/rsif.2008.0484.
31. Cetinay H, Soltan S, Kuipers FA, Zussman G, Van Mieghem P. Comparing the effects of failures in power grids under the ac and dc power flow models. IEEE Trans Netw Sci Eng. 2017;5(4):301–12. https://doi.org/10.1109/TNSE.2017.2763746.
32. Koç Y, Verma T, Araujo NA, Warnier M. Matcasc: a tool to analyse cascading line outages in power grids. In: 2013 IEEE International Workshop on Inteligent Energy Systems (IWIES). IEEE; 2013. pp. 143–8. https://doi.org/10.1109/IWIES.2013.6698576.
33. Christie R. Power systems test case archive. Electrical Engineering Dept, University of Washington. 2000. p. 108 [cited 2024 Feb 18]. https://www2.ee.washington.edu/research/pstca/.
34. Rocchetta R, Patelli E. Assessment of power grid vulnerabilities accounting for stochastic loads and model imprecision. Int J Electr Power Energy Syst. 2018;98:219–32. https://doi.org/10.1016/j.ijepes.2017.11.047.
35. Xiao F, McCalley JD. Power system risk assessment and control in a multiobjective framework. IEEE Trans Power Syst. 2008;24(1):78–85. https://doi.org/10.1109/TPWRS.2008.2004823.
36. Zimmerman RD, Murillo-Sánchez CE, Thomas RJ. MATPOWER: steady-state operations, planning, and analysis tools for power systems research and education. IEEE Trans Power Syst. 2010;26(1):12–9. https://doi.org/10.1109/TPWRS.2010.2051168.
37. Cascading failures simulator in power grids. 2017 [cited 2023 Nov 8]. https://github.com/TUDelftNAS/AC-Cascade-Sim.
38. Cetinay H, Soltan S, Kuipers FA, Zussman G, Van Mieghem P. Analyzing cascading failures in power grids under the AC and DC power flow models. ACM SIGMETRICS Perform Eval Rev. 2018;45(3):198–203. https://doi.org/10.1145/3199524.3199559.
39. Secretaría de Energía. Programa de Desarrollo Del Sistema Eléctrico Nacional (PRODESEN 2019–2033). México DF. 2019 [cited 2022 Oct 16]. https://www.gob.mx/sener/articulos/prodesen-2019-2033-221654.

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